

Individual Contribution to the Project

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My Contributions

- **Project planning:** Participated in brainstorming discussions and explored the dataset to support the project's initial direction.
- **Multilingual dataset preparation:** Researched Tamil code-mixed translation methods, translated the dataset into Tamil and Tanglish, validated sentiment labels, and revised the Tamil translation strategy.
- **Model development:** Analysed tokenizers, built a runnable TF-IDF baseline pipeline, completed the initial transformer setup, and fine-tuned the classifier while exploring suitable model options.
- **OCR development:** Worked on initial OCR model experiments, finalised the model using augmented training images, and established an end-to-end OCR pipeline.
- **OCR masking:** Implemented OCR masking as part of the image-processing workflow.
- **Model hosting:** Hosted the model on Hugging Face and incorporated confidence scores into prediction outputs.
- **Application improvements:** Contributed to backend optimisation and refined the frontend user interface.
- **Testing and reporting:** Supported system testing, organised testing materials, and helped arrange the project report.
- **Evaluation presentations:** Prepared the slides for both the mid-evaluation and final-evaluation presentations.

Challenges and How I Overcame Them

- **Preserving meaning and sentiment across languages:** Addressed Tamil and Tanglish translation difficulties by reviewing translated text, validating sentiment labels, and revising the translation approach.
- **Processing code-mixed text:** Addressed spelling and language-representation differences through tokenizer analysis, a TF-IDF baseline, and transformer model experimentation and fine-tuning.
- **Handling variations in OCR input:** Used augmented training images and refined the OCR pipeline to address differences in image quality and text appearance.
- **Communicating the project clearly:** Organised the evaluation slides and report around the project objectives, methodology, implementation, and outcomes to present the work coherently.